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Untargeted metabolomics reveals key steroidal glycoalkaloids and associated metabolites in crude and C18-purified extracts of potato leaves

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Abstract

Background Potato leaves are an abundant by-product of potato production. Although they contain steroidal glycoalkaloids (SGAs) and other bioactive metabolites, the chemical composition of SGA-enriched leaf extracts and the impact of purification steps remains insufficiently described. Here, we aimed to profile and compare a crude SGA-enriched extract and a C18 solid-phase extraction (SPE)-purified fraction obtained from potato leaves.

Results LC–MS/MS-based untargeted metabolomics annotated 1,761 metabolites across the two extracts, including terpenoids (6.94%), alkaloids (4.90%), and steroids (1.22%). The dominant SGAs were solanidine, α-chaconine, and α-solanine. After data filtering for statistical analysis, 1,760 metabolites were retained; 800 met the differential screening criteria (VIP > 1 and |log2FC| ≥ 1) in purified extract vs. crude extract, whereas 960 did not. The metabolites most affected by C18 purification were mainly organic acids and derivatives, benzene and substituted derivatives, amino acids and derivatives, and glycerophospholipids. In contrast, the major SGAs did not meet the differential screening criteria, suggesting that C18 SPE preserved the core SGA constituents.

Conclusions C18 purification substantially changed a subset of polar co-extracted metabolites but largely maintained the major SGAs in potato leaf extracts. These results provide a metabolite catalog for SGA-enriched potato leaf extracts and a chemical basis for optimizing extraction and purification strategies. Further work should optimize SGA enrichment and evaluate bioactivity and safety to support potential agricultural applications.

Keywords Potato leaves, Steroidal glycoalkaloids, C18 solid-phase extraction, LC–MS/MS, Untargeted metabolomics, Terpenoids

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Introduction

Potato (Solanum tuberosum L.) is an important food and cash crop worldwide. Large amounts of aboveground biomass (leaves and stems) are generated during production, yet much of this material is discarded, resulting in underutilization of plant resources and potential environmental impacts. Potato leaves are known to contain diverse compounds, such as proteins, alkaloids, phenolic acids, and flavonoids [1–4]. However, the chemical composition of SGA-enriched leaf extracts and the identity of co-extracted metabolites remain incompletely characterized, limiting the rational development of value-added applications for this by-product. Plants produce many metabolites with bactericidal, fungicidal, insecticidal, and nematocidal properties to defend against pests and pathogens [5], and plant-derived products are increasingly considered components of sustainable crop management [6–8].

Steroidal glycoalkaloids (SGAs) are nitrogen-containing steroidal secondary metabolites in Solanum species with diverse biological activities [9–11]. In potato, SGAs are concentrated in aerial tissues and in the tuber skin; α-solanine and α-chaconine typically account for the majority of total SGAs in tubers [12]. Because SGAs contribute to plant defense against insects and diseases [13, 14], they have attracted interest as potential leads for plant-based crop-protection agents [15]. For example, potato leaf extracts or purified α-solanine have been reported to affect oxidative stress markers and fitness traits in insect pests [16], and leaf extracts have been evaluated as alternatives to synthetic fungicides against post-harvest pathogens [17, 18]. Beyond SGAs, terpenoids and other metabolite classes also contribute to the leaf chemical profile and may influence extract functionality [19, 20]. Advances in LC–MS/MS-based metabolomics now enable broad characterization of complex plant extracts. SGA biosynthesis shares precursors and intermediates with the terpenoid and phytosterol pathways [21], suggesting that SGA-enriched extracts may co-extract related metabolites. Nevertheless, comparative metabolomics of crude versus purified SGA-enriched extracts from potato leaves has rarely been reported. In this study, we applied untargeted metabolomics to (i) catalog metabolites detected in a crude extract and a C18 SPE-purified fraction from potato leaves, (ii) identify the major SGAs present, and (iii) assess how the purification step reshapes the overall metabolite profile. These data provide a foundation for future optimization and utilization of potato leaf by-products.

Materials and methods

Plant materials

The potato cultivar 'Jizhangshu No. 12' (*Solanum tuberosum* L.) was grown at the Potato Experimental Base of the Agricultural and Forestry Science Research Institute in Ulanqab, Inner Mongolia, China. Plants were sown on 3 May 2024 and managed under routine field irrigation. Leaves were harvested at the late growth stage (20 August 2024). Upper leaves without visible pest or disease symptoms and without yellowing were collected, immediately weighed, dried at 70 °C to constant weight, and ground to a fine powder. The powder was stored in sealed bags at room temperature (20 °C) until extraction. Three independent biological replicates were prepared ($n = 3$) for each extract type.

Preparation of steroidal glycoalkaloid-enriched extracts

SGA-enriched crude extracts were prepared following Nie et al. [22], with minor modifications. Briefly, 5 g of potato leaf powder was mixed with 75 mg cellulase (1.5%, Sangon, China) in a sealed container and extracted with 40 mL acidic aqueous solution (0.02 mol/L HCl, pH 5). The mixture was vortex-mixed and sonicated (300 W, KQ-300DE, Kunshan, China) in a 45 °C water bath for 30 min, then incubated for enzymatic hydrolysis at 45 °C for 1.5 h. Enzyme activity was terminated by heating to 95 °C. Ethanol was added to reach 70% (v/v) (140 mL ethanol to a final volume of 200 mL), and the mixture was left to stand for 18 h. The extract was vacuum-filtered and concentrated using a rotary evaporator (SB-1200, Anting, China). The residue was dissolved in 30 mL 5% H2SO4 and filtered, then the filtrate pH was adjusted to 10.0–10.5 using concentrated ammonia solution (25–28% w/w). After overnight precipitation at 4 °C, the sample was centrifuged (4,000 rpm, 20 min) and the supernatant discarded. The precipitate was washed repeatedly with 1% ammonia water until the wash solution was clear. The crude precipitate was dissolved in 5 mL methanol and decolorized with activated carbon to obtain the crude methanol extract for LC–MS/MS analysis.

Purification of crude SGA extracts

Crude extract was purified using a C18 SPE extraction cartridge (Bond Elut C18, 100 mg, 6 mL, 40 μm, Agilent). The cartridge was conditioned with 5 mL methanol and equilibrated with 5 mL distilled water. 6 mL of the crude extract were loaded, and the flow-through was discarded. The cartridge was washed with 5 mL distilled water. The retained fraction was then eluted with 5 mL methanol. The methanol eluate was collected under vacuum and designated as the purified fraction for LC–MS/MS analysis.

LC–MS/MS sample preparation and quality control

Sample preparation

Aliquots of crude extract and purified extract were stored at − 80 °C and thawed on ice prior to analysis. For each sample, 100 μL of extract was mixed with 100 μL of 70%

methanol containing internal standards and vortexed for 30 s. The mixture was incubated on ice for 15 min and centrifuged at 12,000 rpm for 15 min at 4 °C. The supernatant was filtered through a 0.22 μm membrane and transferred to LC–MS vials for analysis. A pooled quality-control sample was prepared by mixing equal aliquots of all samples and analyzed periodically to monitor instrument stability. Three biological replicates (n = 3) were analyzed for each extract type in both positive and negative ion modes.

HPLC conditions

LC–MS/MS analysis was performed following Yang et al. [23], with minor modifications. Chromatographic separation was carried out on a Waters ACQUITY Premier HSS T3 column (2.1 × 100 mm, 1.8 μm) maintained at 40 °C with a flow rate of 0.4 mL/min and an injection volume of 5 μL. Mobile phase A was water containing 0.1% (v/v) formic acid and mobile phase B was acetonitrile containing 0.1% (v/v) formic acid. The gradient is shown in Table 1. Each sample was analyzed in both positive and negative electrospray ionization modes using the same chromatographic conditions.

MS conditions

Mass spectrometry was operated in information-dependent acquisition mode controlled by Analyst TF software (version 1.7.1, Sciex) [23]. Ion source parameters were: source temperature 500 °C; ion spray voltage 5,500 V (positive) and – 4,500 V (negative); curtain gas 25 psi; ion source gas 1 and gas 2 both 50 psi; declustering potential 80 V (positive) and – 80 V (negative). TOF-MS data were acquired over m/z 60–1200 and product ion spectra over m/z 25–1200 using collision energy 30 V (positive) or – 30 V (negative) with a collision energy spread of 15 V.

Data preprocessing

Raw data files were converted to mzXML format using ProteoWizard and processed in R for peak detection, alignment, and retention-time correction. Features with > 50% missing values were excluded; remaining missing values were imputed using k-nearest neighbors. Peak intensities were normalized using support vector regression based on quality-control samples. Metabolite annotation was performed by MetWare Biotechnology (Wuhan, China) by matching accurate mass and MS/MS spectra against the MWDB and public databases. High-confidence annotations (ID score ≥ 0.5) with acceptable quality-control reproducibility (CV < 0.5) were retained for downstream analysis.

Table 1 Model parameters

Group	R^2X(cum)	R^2Y(cum)	Q^2(cum)	pre	ort
Crude extract vs. Purified extract	0.71	0.99	0.93	1.00	1.00

Data analysis

Principal component analysis

Unsupervised principal component analysis (PCA) was performed via the statistics function prcomp in R (www.r-project.org; R version 4.1.2). The data were subjected to unit-variance scaling before unsupervised PCA.

Hierarchical cluster analysis

Hierarchical cluster analysis (HCA) of samples and metabolites was performed in R (version 4.1.2) using the ComplexHeatmap package. Prior to clustering, metabolite peak intensities were normalized and unit-variance scaled. Sample-to-sample similarity was assessed by Pearson correlation coefficients computed with the cor function in R. The resulting clustering and correlation matrices were used for downstream visualization.

Selection of differentially abundant metabolites

For two-group comparisons (purified extract vs. crude extract), differentially abundant metabolites were screened based on variable importance in projection (VIP > 1) from the OPLS-DA model and absolute $\log_2$ fold change ($|\log_2 FC| \geq 1$). Prior to modelling, peak intensities were $\log_2$-transformed and mean-centered. Model overfitting was assessed by 200 permutation tests. Where applicable, univariate p-values were calculated using two-sided Student's t-tests on $\log_2$-transformed data, and p-values were adjusted for multiple testing using the Benjamini–Hochberg procedure.

Metabolite annotation

Metabolite identification was performed by MetWare Biotechnology (Wuhan, China) by matching accurate mass and MS/MS spectra against the custom-built MWDB database and publicly available metabolite databases. The analysis software used for substance identification was developed by MetWare Biotechnology. According to the priority of data analysis, searching was performed against the local high-resolution database built by the company (MS2-search-local), followed by public online libraries including Metlin (http://metlin.scripps.edu/index.php), HMDB (https://hmdb.ca/), MoNA (https://mona.fiehnlab.ucdavis.edu/), and MassBank (http://www.massbank.jp/) (MS2-search-DB), and finally an in silico prediction library based on machine learning (MS2-In silico). Metabolite abundance was quantified as normalized peak intensities (integrated peak areas) obtained after peak detection/alignment and quality-control-based normalization and was used for relative comparisons between groups. For substances that were

not identified at the second level, an MS1-based candidate was provided as a reference on the basis of the Q1 error (MS1).

Statistical analyses

Details of software and parameters used for plotting are provided in the supplementary materials (Table S1). Statistical analyses were performed in R (version 4.1.2). Normality of $\log_2$-transformed metabolite intensities was assessed using the Shapiro–Wilk test to guide the choice of univariate test. Two-group comparisons were conducted using a two-sided Student's t-test when the normality assumption was approximately met; otherwise, the Wilcoxon rank-sum test was applied. For multiple testing, *p*-values were adjusted using the Benjamini–Hochberg procedure when applicable. Graphs were prepared in R and Microsoft Excel.

Results

Quality control analysis of crude and purified SGA extracts

To characterize the composition of SGA-enriched extracts from potato leaves, we performed LC–MS/MS-based untargeted metabolomics on the crude extract and the C18-purified fraction. Total ion chromatograms of pooled quality-control injections showed strong overlap in both positive and negative ion modes (Figure S1), indicating good retention-time and signal stability. Blank injections showed no obvious internal-standard carryover (Figure S2). Pearson correlation analysis among quality-control injections further demonstrated high reproducibility across the analytical batch (Fig. 1).

PCA analysis of crude and purified SGA extracts

In the PCA model, the first two principal components (PC1 and PC2) explained 51.81% and 23.15% of the total variance, respectively (74.96% cumulatively). Crude extract and purified extract samples were clearly separated in the PCA score plot (Fig. 2a), indicating distinct metabolite profiles. Quality-control-based coefficient of variation (CV) analysis showed that >85% of detected features had CV < 0.5 (Fig. 2b), supporting acceptable analytical stability.

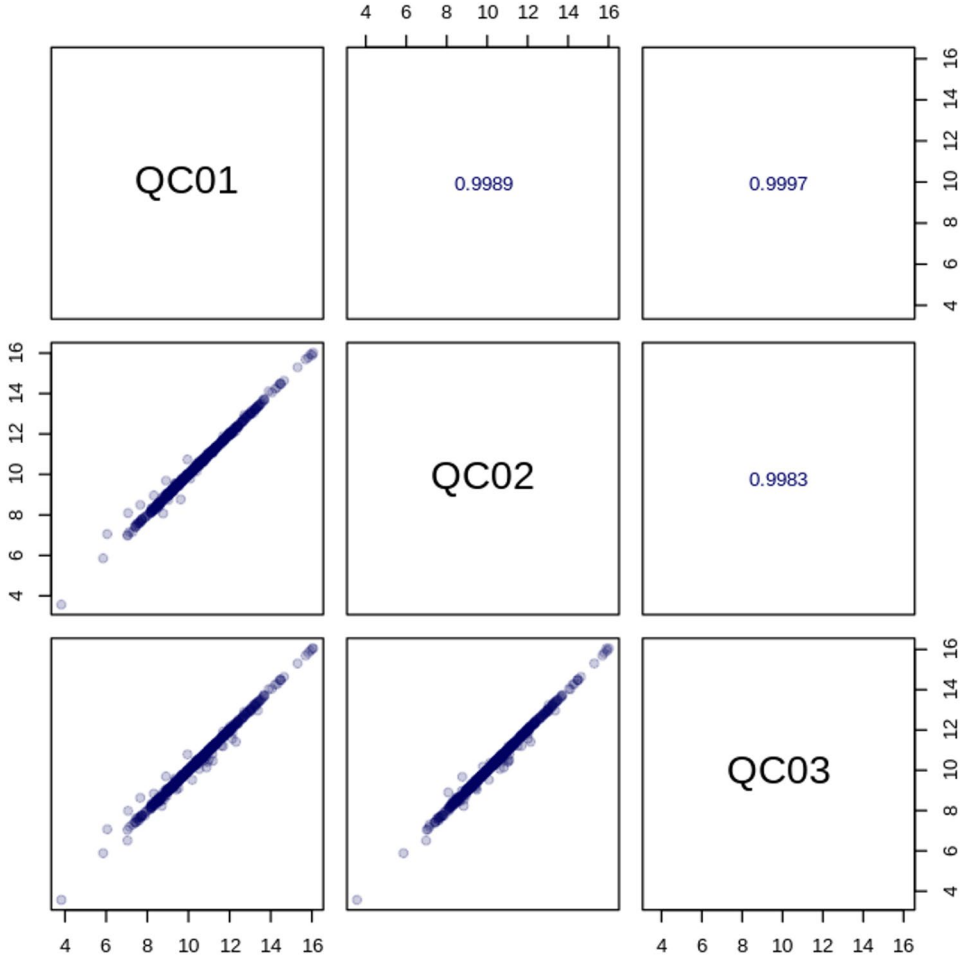


Fig. 1 Pearson correlation analysis of quality-control injections (QC01–QC03) demonstrating high reproducibility across the LC–MS/MS run

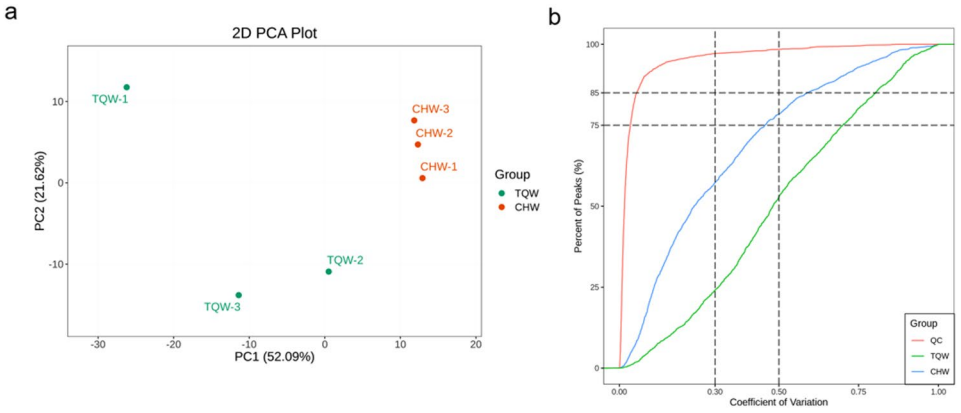


Fig. 2 **a** PCA score plot of metabolomic profiles for crude extract and purified extract ($n = 3$ biological replicates each). **b** CV distribution in quality-control samples; the proportion of features with CV < 0.5 exceeded 85%. TQW is crude extract. CHW is purified extract

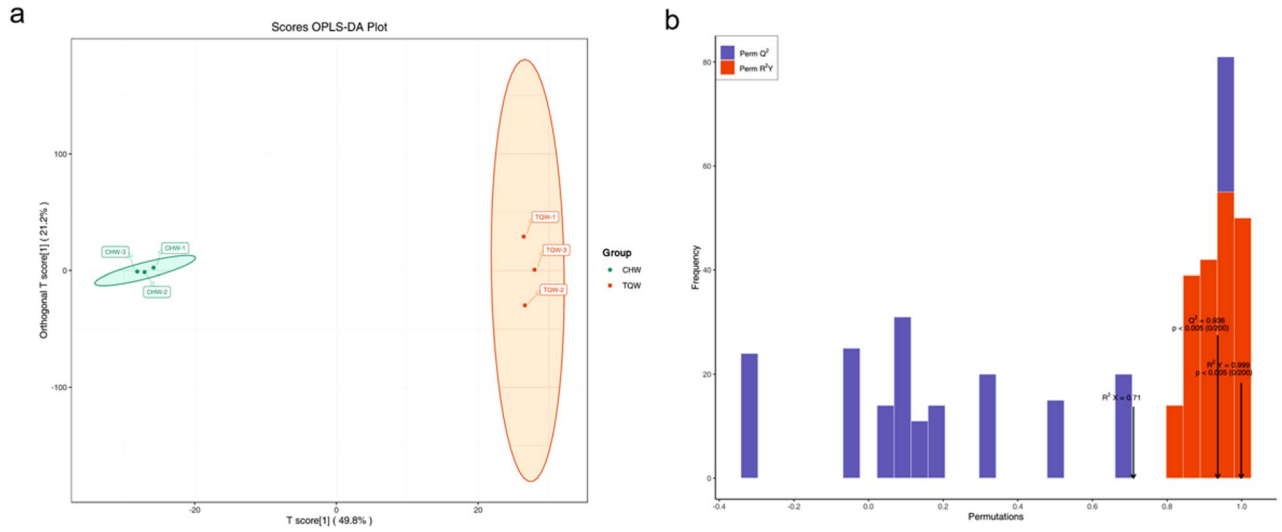


Fig. 3 OPLS-DA modelling of crude extract and purified extract metabolomic profiles. **a** OPLS-DA score plot ($n = 3$ biological replicates each). **b** Permutation test ($n = 200$) for assessing model overfitting. TQW is crude extract. CHW is purified extract

OPLS-DA analysis of crude and purified SGA extracts

To further assess differences between crude extract and purified extract, a supervised OPLS-DA model was built. The two extract types formed distinct clusters in the OPLS-DA score plot (Fig. 3a). Model statistics (R^2Y and Q^2) indicated good fit and predictive performance, and permutation testing (200 permutations) suggested that the model was not overfitted (Fig. 3b, Table 1).

Screening of differentially abundant metabolites

A total of 1,761 metabolites were annotated across all samples (Table S2), including 1,316 features detected in positive ion mode and 445 in negative ion mode. The largest classes were amino acids and derivatives (21.22%), organic acids (10.61%), lipids (8.98%), benzene and substituted derivatives (8.37%), terpenoids (6.94%), alkaloids (4.90%) and steroids (1.22%) (Fig. 4a). Among the top 10 metabolites by mean relative intensity were several lipids and amides as well as the SGA-related compounds solanidine and α-chaconine, highlighting that the extraction procedure enriched SGAs within a complex metabolite matrix.

To visually discern inter-group distinctions and pinpoint differential metabolites, orthogonal partial least squares discriminant analysis (OPLS-DA) was used as a supervised multivariate approach (Fig. 4b).

After filtering for missing values and quality control reproducibility, 1,760 metabolites were retained for statistical comparison. Using VIP > 1 and $|\log_2 FC| \geq 1$ as screening criteria, 800 metabolites were identified as differentially abundant in purified extract vs. crude extract (Table S3), including 162 higher and 638 lower in purified extract (Fig. 4c). Hierarchical clustering of these metabolites showed clear separation between purified extract and crude extract (Fig. 5). The most strongly increased metabolites in purified extract ($\log_2 FC > 6$) included

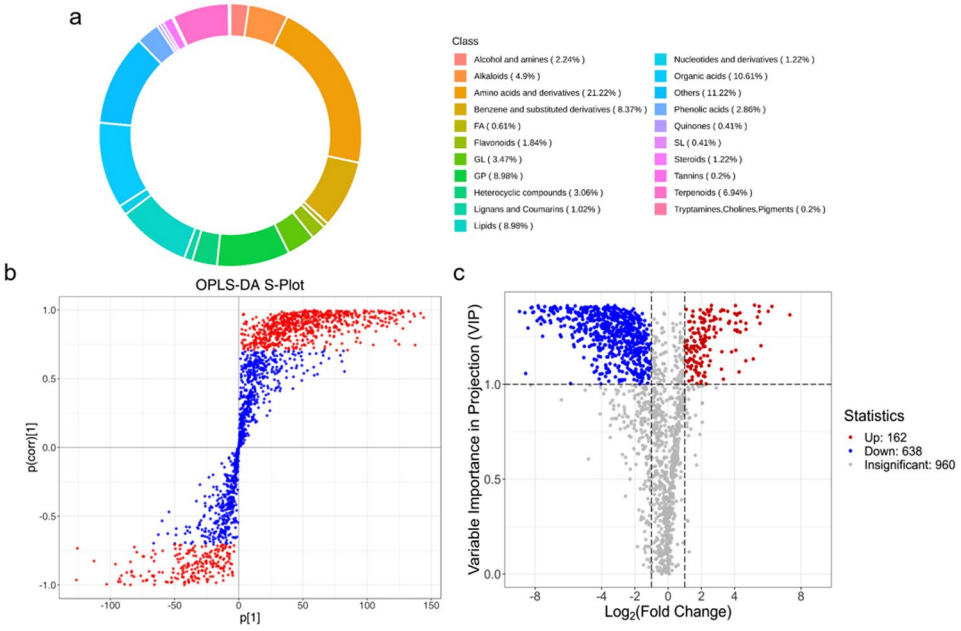


Fig. 4 Metabolite profiling and differential screening between purified extract and crude extract. **a** Composition of annotated metabolite classes. **b** OPLS-DA S-plot showing covariance and correlation of variables; metabolites with VIP > 1 are highlighted. **c** VIP–fold-change plot (log_2FC vs. VIP) for purified extract vs. crude extract, points are coloured according to the screening criteria (VIP > 1 and $|log_2FC| \geq 1$)

2-hydroxyvaleric acid, 2-phenylbutyramide, and 1-palmitoyl-2-oleoyl-sn-glycero-3-phosphate, whereas several amino acids/peptides were among the most decreased metabolites.

In contrast, 960 metabolites did not meet the differential criteria between crude extract and purified extract (Table S4). These included 48 alkaloids (including steroidal alkaloids such as α-solanine, α-chaconine and solanidine), 37 terpenoids, and 9 steroids. Overall, these results indicate that C18 SPE purification altered a subset of metabolites while leaving many alkaloid and steroid-related constituents largely unchanged.

Relative abundances of metabolite classes in crude and SPE purified extracts

Among the annotated metabolites, 84 alkaloids were identified (Table S5), including seven steroidal alkaloids (e.g. solanidine, α-chaconine, and α-solanine), seven terpenoid alkaloids, six quinoline/isoquinoline alkaloids, and other alkaloids. In the purified extract vs. crude extract comparison, some non-SGA alkaloids showed increased or decreased abundance after purification; however, 48 alkaloids did not meet the differential criteria. Notably, the dominant steroidal alkaloids (solanidine, α-chaconine, and α-solanine) showed high mean normalized peak intensity in both extracts (Fig. 6a) and did not meet the differential criteria, indicating that C18 SPE largely retained these major SGAs.

Terpenoids represented 6.94% of annotated metabolites (Fig. 4a). In total, 74 terpenoid-related metabolites were annotated (Table S6), including diterpenes, sesquiterpenes, triterpenes and saponins. Thirty-eight terpenoids met the differential criteria in purified extract vs. crude extract, whereas 36 did not. Several terpenoids showed relatively high mean intensities in both extracts (Fig. 6b), suggesting that terpenoids co-extract with SGAs and contribute to the overall chemical profile of potato leaves.

Steroid-related metabolites (excluding steroidal alkaloids) accounted for 1.22% of annotated metabolites (Fig. 4a). Twenty-one steroids were annotated (Table S7), including steroidal saponins and cardiac glycosides. In purified extract vs. crude extract, 12 steroids met the differential criteria and nine did not. Some individual steroids (e.g. capsicoside C) showed higher mean relative intensities in crude extract than in purified extract (Fig. 6c), suggesting partial loss during C18 purification.

Discussion

Compared with previous LC–MS profiling of potato leaves, which described a chemically diverse leaf metabolome including organic acids, amino acids, phenolics, and other specialized metabolites [4], our untargeted analysis specifically focused on crude and C18-purified SGA-enriched extracts and thus provides a more application-oriented view of the metabolite background accompanying SGA enrichment. Within these extracts, α-chaconine and α-solanine remained the most representative potato SGAs, consistent with earlier potato leaf metabolomics and comprehensive reviews of potato glycoalkaloids [4, 24]. The detection of solanidine, the

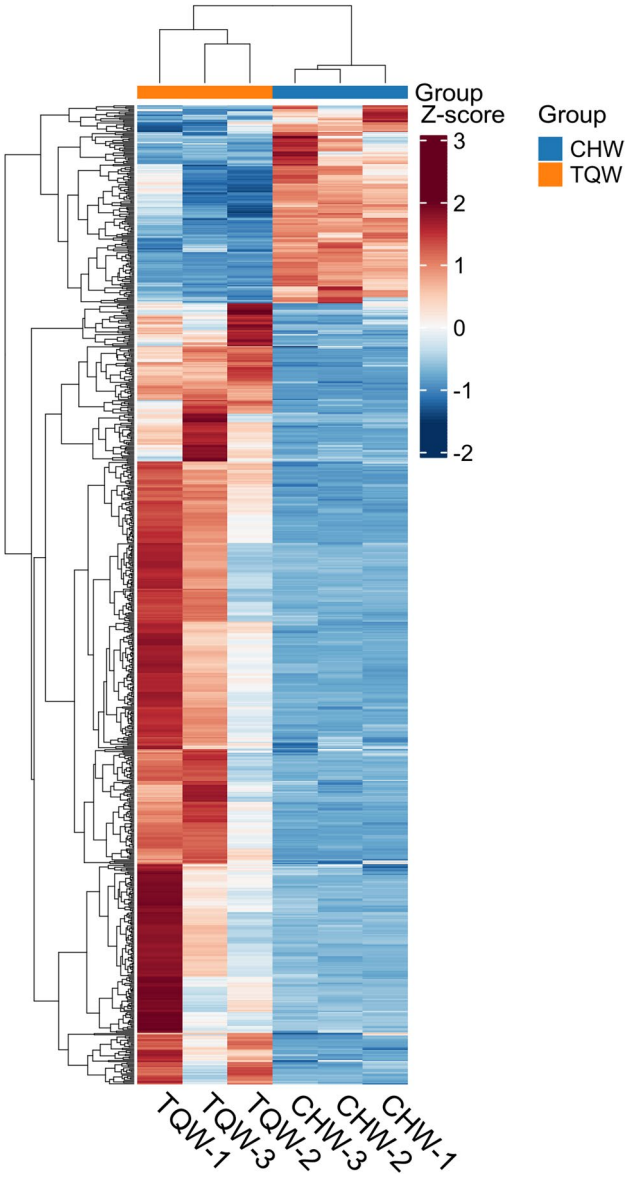


Fig. 5 Heatmap and hierarchical clustering of differentially abundant metabolites (screened by VIP and $|\log_2 FC|$ criteria) in crude extract and purified extract. TQW is crude extract. CHW is purified extract

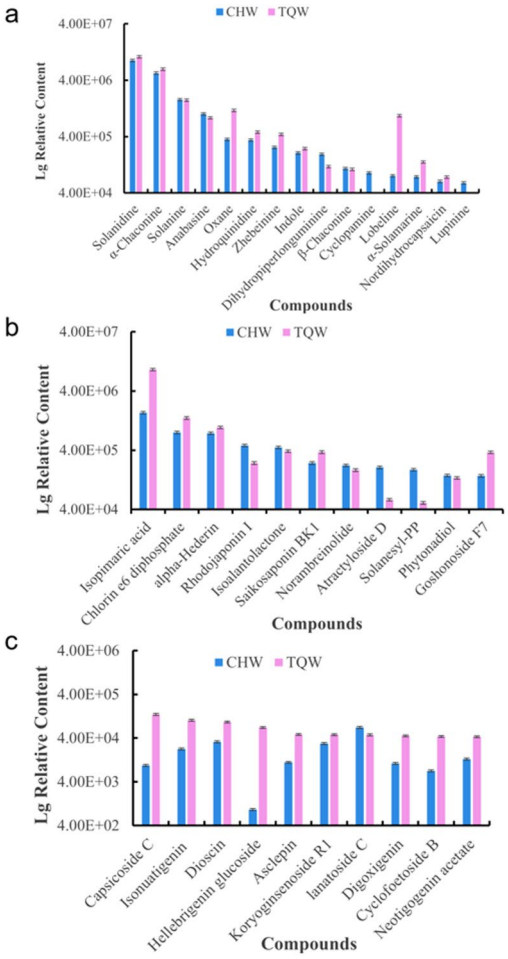


Fig. 6 Mean normalized peak intensities (integrated peak area units) of (**a**) alkaloids, (**b**) terpenoids, and (**c**) steroids in crude extracts and C18-purified extracts. TQW denotes crude extract and CHW denotes purified extract

aglycone backbone of potato glycoalkaloids, further supports that the enriched fractions captured the core SGA-related chemical space of potato leaves. These observations are also consistent with the current understanding of SGA metabolism in Solanum. SGA accumulation is linked to sterol- and isoprenoid-associated biosynthetic pathways, and cultivar-dependent differences have been reported under glycoalkaloid-inducing conditions [21, 25]. Our dataset therefore extends earlier pathway-oriented studies by showing, at the extract level, that the dominant SGA-related constituents remain prominent even against a broad background of co-extracted metabolites.

The comparison between purified extract and crude extract indicates that C18 SPE mainly reduced co-extracted polar metabolites, particularly organic acids and derivatives, amino acids and derivatives, benzene and substituted derivatives, and glycerophospholipids, while the major SGAs did not meet the differential screening criteria. Because C18 retention favors relatively hydrophobic constituents, this pattern is chemically plausible and suggests that the SPE step simplified matrix composition without causing an obvious loss of the dominant SGAs under our untargeted screening criteria. In practical terms, this matrix simplification may facilitate downstream isolation, targeted quantification, or biological testing of SGA-enriched fractions, although absolute recoveries were not measured here.

From an application perspective, potato leaf residues are abundant but underused by-products. The prominence of α-chaconine, α-solanine, and solanidine is relevant because potato glycoalkaloids have been associated with defense against insects and pathogens and have

been discussed as bioactive constituents with agricultural or pharmacological potential [13–16, 24, 26]. Our results do not establish bioactivity directly, but they provide a chemically resolved basis for prioritizing which metabolites and extract fractions should be taken forward for crop-protection assays or other value-added uses.

Several limitations should be considered when interpreting these data. Metabolite identities are putative and based on database matching, so structural confirmation with authentic standards will be required, especially for isomers. The present study reports normalized peak intensities for relative comparisons; we did not convert these signals to semi-quantitative concentrations relative to the internal standards because ionization responses differ markedly among metabolite classes in untargeted LC–MS/MS datasets. Future targeted analyses using authentic standards should quantify α-chaconine, α-solanine, solanidine, and other priority SGAs in absolute or semi-quantitative terms. In addition, only one cultivar and one harvest time were analyzed, and previous work indicates that glycoalkaloid metabolism can vary among genotypes and treatments [25]. Drying at 70 °C may also have affected some co-extracted heat-sensitive metabolites, although the extent of any effect on individual SGAs was not assessed here. Bioactivity and safety evaluations were beyond the scope of this study and remain necessary before specific practical applications can be recommended.

Conclusions

In conclusion, this study extends previous potato leaf metabolomics by comparing crude and C18-purified SGA-enriched extracts rather than whole-leaf extracts alone. We annotated 1,761 metabolites and identified solanidine, α-chaconine, and α-solanine as the dominant SGA-related constituents. The C18 SPE step reduced many co-extracted polar metabolites while largely preserving the major SGAs detected by untargeted LC–MS/MS, providing a useful analytical reference for future optimization of SGA enrichment, targeted quantification, and evaluation of potato leaf by-products for agricultural applications.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12870-026-08606-7.

Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Supplementary Material 4.

Supplementary Material 5.

Supplementary Material 6.

Supplementary Material 7.

Supplementary Material 8.

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Not applicable.

Authors' contributions
All the listed authors contributed to this study and agreed to publish it. Lizhen Nie conceived the project. Yuru Lv wrote the main manuscript text and prepared all the figures and tables. Yue Chang and Yating Hao designed the experiments and analyzed the data. Yating Hao, Zhicheng Zhang, Rui Xie, Jie Feng collected plant material. Ruigang Wang, Xianmei Xi, Yongyu Fang and Lizhen Nie revised and supervised the manuscript. All authors have read and agreed to the published version of the manuscript.

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Data availability
All data generated or during this study are included in this published article and its supplementary information files. Additional data are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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